Radio Frequency Engineering Project

Design and Optimization of
A 7th-Order Microstrip Low-Pass Filter

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1. Introduction & Filter Specifications

Modern microwave communication systems require high-performance filtering structures to eliminate out-of-band spurious signals and harmonic interference. The goal of this project is to design, model, and optimize a physical planar microstrip low-pass filter utilizing full-wave 3D electromagnetic simulation software (CST Microwave Studio).

The operational design requirements specified for this filter are:

- **Filter Order (N):** 7th-Order
- **Passband Characteristic:** Equal-ripple Chebyshev response
- **Cutoff Frequency (f_c):** 5.0 GHz
- **System Characteristic Impedance (Z_0):** 50 Ω
- **Substrate Material:** Lossless FR4 (
- **Substrate Thickness (h):** 2.0 mm

2. Theoretical Hand Analysis & Filter Synthesis

The translation of a lumped-element low-pass ladder network into a physically realizable microstrip format is achieved by sequentially executing three mathematical operations: low-pass prototype evaluation, Richard's Transformation mapping, and network translation via Kuroda's Identities.

2.1 The Normalized Lumped-Element Prototype

The synthesis originates from a normalized low-pass prototype (g_k parameters) optimized for an equal-ripple Chebyshev profile. For an odd-order filter ($N = 7$), the network exhibits perfect physical and electrical symmetry around the central element (g_4), such that $g_k = g_{8-k}$. The normalized source and load termination impedances are bounded at $g_0 = g_8 = 1$, with the normalized cutoff frequency established at $\Omega_c = 1$. The prototype network alternates between series inductors ($L_k = g_k$) and shunt capacitors ($C_k = g_k$).

2.2 Richard's Transformation

Because lumped inductors and capacitors suffer from self-resonance and radiation losses at microwave frequencies, Richard's Transformation is employed to map the lumped complex frequency domain ($s = j\omega$) to the distributed commensurate transmission line domain ($S = j\Omega$) via the tan-mapping variable:

$$\Omega = \tan(\beta l) = \tan\left(\frac{\omega l}{v_p}\right)$$

Setting the normalized cutoff frequency boundary to $\Omega_c = 1$ at the physical target cutoff ($f_c = 5.0$ GHz) isolates the electrical length ($\theta = \beta l$) of the distributed sections:

$$1 = \tan(\beta l) \Rightarrow \theta = \beta l = \arctan(1) = 45^\circ = \frac{\pi}{4} \text{ rad}$$

This establishes that every transmission line segment in the commensurate network must possess an electrical length of exactly 45° (corresponding to an eighth-wavelength segment, $\lambda_g/8$) at the cutoff frequency. Under this mapping:

- **Lumped Series Inductors (L):** Map to **Short-Circuited Series Stubs** with a characteristic impedance $Z = L$.
- **Lumped Shunt Capacitors (C):** Map to **Open-Circuited Shunt Stubs** with a characteristic impedance $Z = \frac{1}{C}$.

2.3 Step-by-Step Kuroda Transformation Derivations

Implementing series short-circuited stubs is highly impractical in planar microstrip circuits because cutting a series slot into the center strip disrupts the ground plane isolation and introduces severe uncontrolled parasitics. To transform the network into an all-shunt configuration, redundant transmission line segments known as **Unit Elements (UEs)** are introduced at the input and output boundaries. These UEs possess an electrical length of $\theta = 45^\circ$ and match the $50\ \Omega$ system impedance ($Z_0 = 1$ in normalized form). They do not alter the magnitude filtering profile but provide the algebraic handles required to apply **Kuroda's Identities**.

The localized transformation matrix relies on the scaling index n^2:

$$n^2 = 1 + \frac{Z_2}{Z_1}$$

Kuroda's identities physically shift the Unit Elements inward, transforming the un-manufacturable short-circuited series stubs into parallel open-circuited shunt stubs positioned entirely on the outer perimeter of the main line. The absolute physical characteristic impedances (Z_{physical}) are subsequently computed by denormalizing the final distributed values against the terminal reference environment via the scaling relation:

$$Z_{\text{physical}} = Z_{\text{normalized}} \times 50$$

2.4 Hand Calculations and Network Schematics Documentation

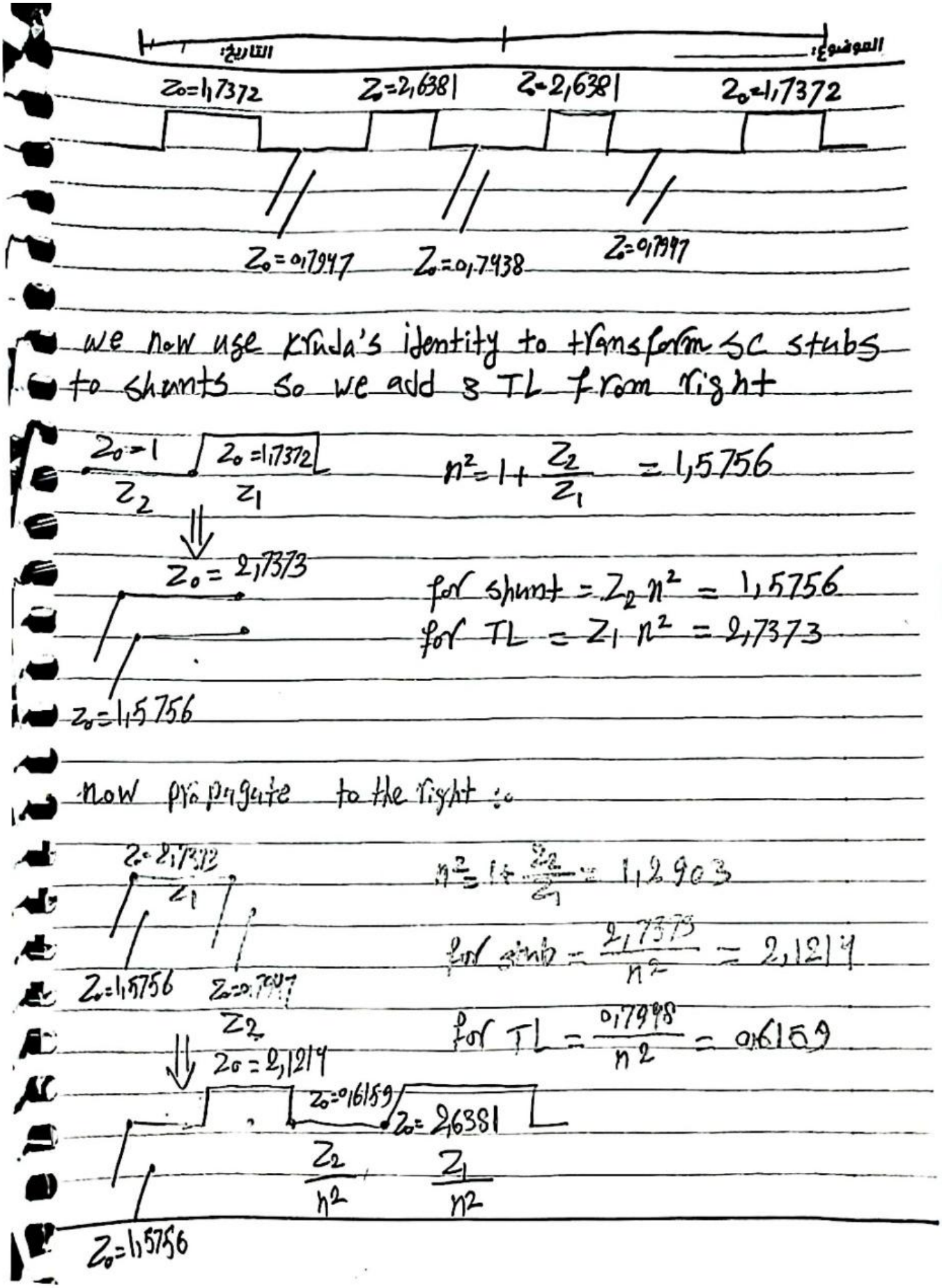


Figure 2.1: Hand Calculations Sheet 1.

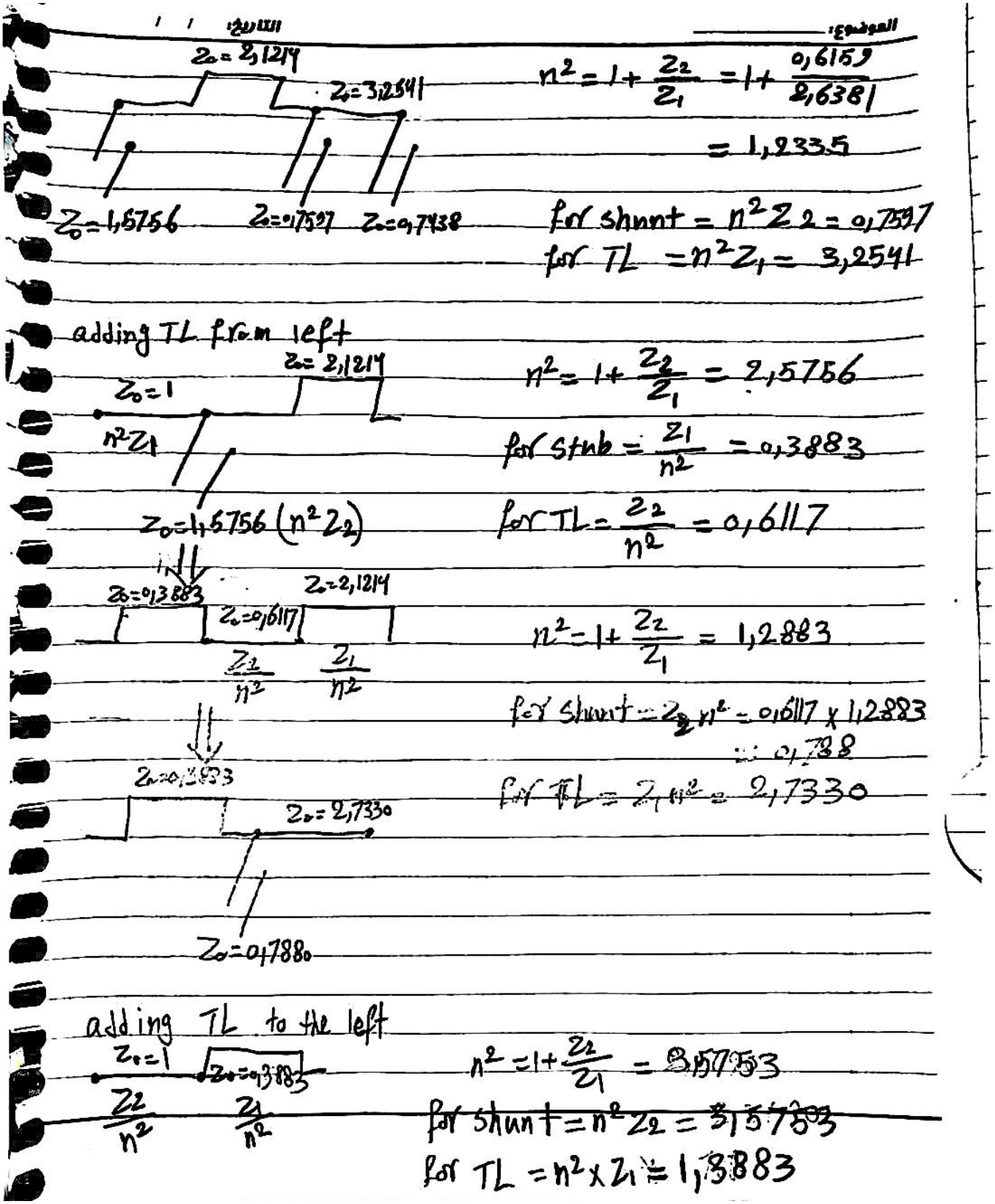


Figure 2.2: Hand Calculations Sheet 2.

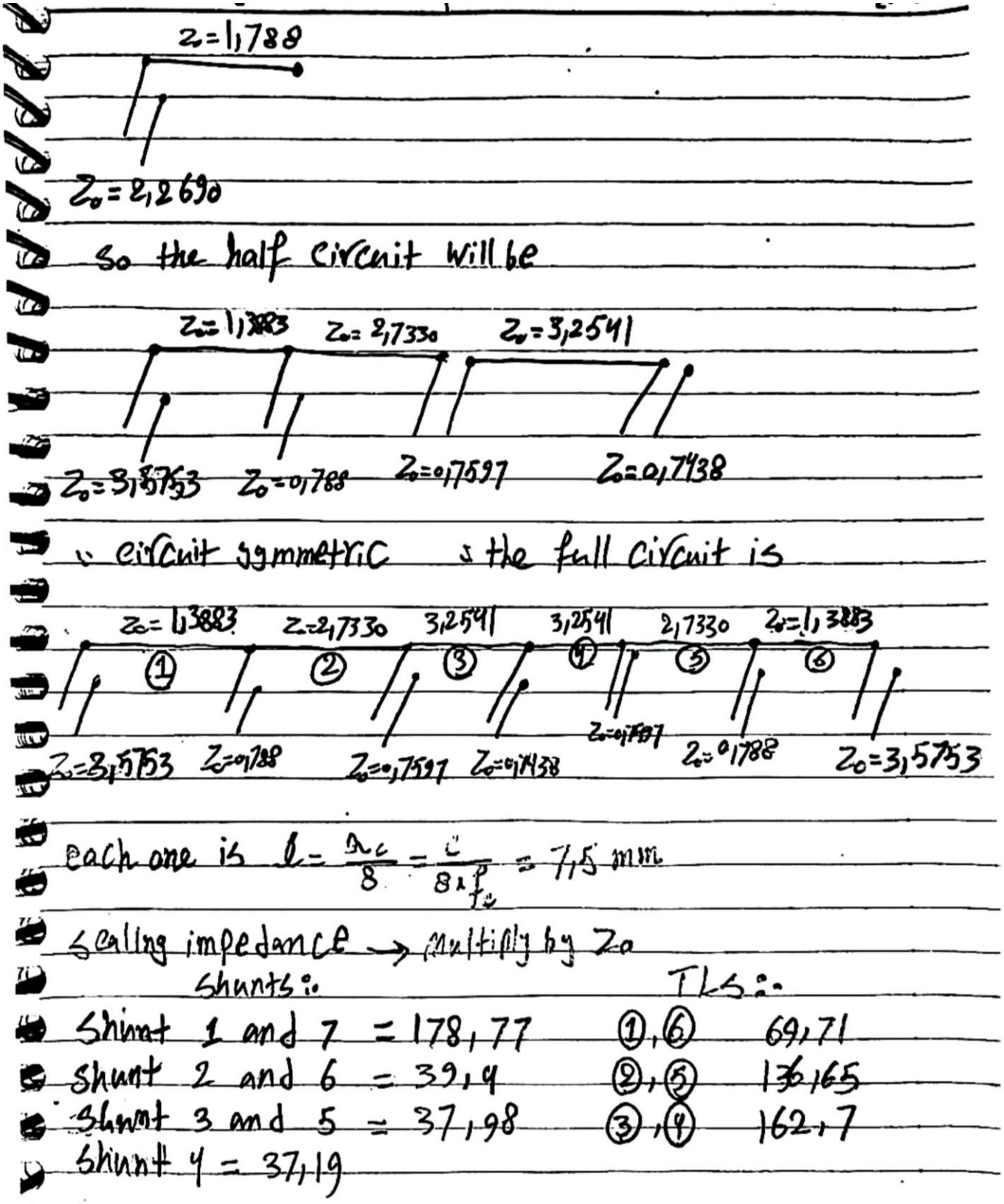


Figure 2.3: Hand Calculations Sheet 3.

2.5 Final Hand-Calculated Target Impedances

The definitive characteristic impedances (Z_0) computed manually for the 7th-order low-pass distributed network are systematically compiled in **Table 2.1**.

Table 2.1: Analytical Impedance Parameter Summary for the 7th-Order Distributed LPF.

Component Reference Name	Distributed Element Classification	Target Impedance (Z0)
Input / Output Feed Lines	50 Ω Terminal Interface Line	$\mathbf{50.00}\ \Omega$
Shunt Stub 1 & 7 (Z_{Stub1})	Outermost Parallel Open-Circuit Shunt Stub	$\mathbf{178.77}\ \Omega$
Unit Element 1 & 6 (Z_{UE1})	High-Impedance Connecting Series Line	$\mathbf{69.41}\ \Omega$
Shunt Stub 2 & 6 (Z_{Stub2})	Intermediate Parallel Open-Circuit Shunt Stub	$\mathbf{39.40}\ \Omega$
Unit Element 2 & 5 (Z_{UE2})	High-Impedance Connecting Series Line	$\mathbf{136.65}\ \Omega$
Shunt Stub 3 & 5 (Z_{Stub3})	Innermost Parallel Open-Circuit Shunt Stub	$\mathbf{37.98}\ \Omega$
Unit Element 3 & 4 (Z_{UE3})	Central Connecting Series Line	$\mathbf{162.70}\ \Omega$
Shunt Stub 4 (Z_{Stub4})	Absolute Center Parallel Open-Circuit Shunt Stub	$\mathbf{37.19}\ \Omega$

3. Microstrip Realization & 3D Geometric Layout

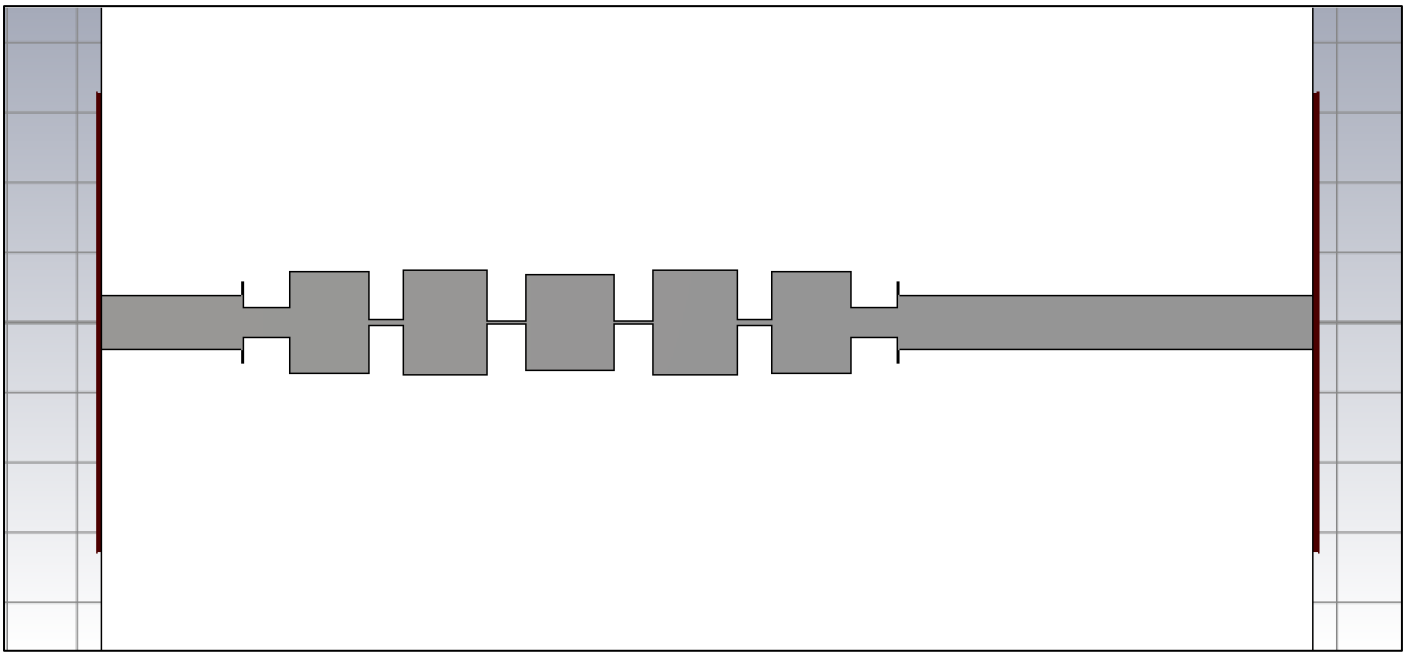


Figure 3: Filter Front view.

3.1 Dielectric Substrate Height Engineering

To map extreme characteristic impedances like 178.77 Ω and 162.70 Ω into a physically realizable trace width, the substrate thickness was deliberately increased to **2.0 mm** on a lossless FR4 board. Based on microstrip field equations, the characteristic impedance scales inversely with the width-to-height aspect ratio:

$$Z_0 \propto \ln\left(\frac{h}{W}\right)$$

By selecting a thick 2.0 mm dielectric slab, the trace width for the extreme 178.77 Ω line expanded to a manageable, highly precise width of **0.1075 mm**.

3.2 Precise 3D Coordinate Mapping Expressions

To construct the 7th-order filter topology in CST without rounding errors, a cumulative, gap-free mathematical coordinate script was implemented using the parameter variables. The circuit baseline originates at the input boundary ($X_{\text{start}} = -L_{\text{sub}}/2$).

The absolute geometric boundaries utilized to model the foundational blocks of the half-circuit are detailed below:

- **Component 0: 50 Ω Input Feed Line (Input_Line)**
  - Xmin: -L_sub/2 | Xmax: -L_sub/2 + L_50
  - Ymin: -W_50/2 | Ymax: W_50/2
  - Zmin: 2 | Zmax: 2 + 0.035 (Standard 35 μm copper thickness)
- **Component 1: Shunt Stub 1 Cross (Stub1_Top & Stub1_Bottom)**
  - Xmin: -L_sub/2 + L_50 | Xmax: -L_sub/2 + L_50 + W_Stub1
  - Ymin (Bottom): -L_Stub1 | Ymax (Bottom): 0
  - Ymin (Top): 0 | Ymax (Top): L_Stub1
- **Component 2: Unit Element 1 Line (UE1)**
  - Xmin: -L_sub/2 + L_50 + W_Stub1 | Xmax: -L_sub/2 + L_50 + W_Stub1 + L_UE1
  - Ymin: -W_UE1/2 | Ymax: W_UE1/2
- **Component 3: Shunt Stub 2 Cross (Stub2_Top & Stub2_Bottom)**
  - Xmin: -L_sub/2 + L_50 + W_Stub1 + L_UE1 | Xmax: -L_sub/2 + L_50 + W_Stub1 + L_UE1 + W_Stub2
  - Ymin (Bottom): -L_Stub2 | Ymax (Bottom): 0
  - Ymin (Top): 0 | Ymax (Top): L_Stub2

(This absolute cumulative expression sequence continues systematically up to the center cross-stub Stub4. The remaining half of the filter is generated via a precise structural mirror reflection across the center plane origin axis: $X_0 = -L_{sub}/2 + L_{50} + W_{Stub1} + L_{UE1} + W_{Stub2} + L_{UE2} + W_{Stub3} + L_{UE3} + W_{Stub4}/2$).

4. Full-Wave Simulation & Optimization Workflow

4.1 The Parasitic Frequency Downward Shift

The fully assembled microstrip model was executed inside the **CST Frequency Domain Solver**. The initial baseline curve revealed an exceptional equal-ripple Chebyshev profile featuring exactly 3 distinct return loss dips in the real, positive passband spectrum. However, the physical cutoff frequency shifted downward, landing at **3.5 GHz instead of the targeted 5.0 GHz**.

This severe frequency detuning is driven by two physical electromagnetic phenomena omitted by ideal transmission line equations:

1. **Open-End Fringing Extensions (Δl):** The electric fields traveling along the open-circuited tuning stubs do not truncate abruptly at the physical copper edge; they expand slightly into the surrounding air. This extra fringing capacitance creates a physical end-effect extension, making every stub behave as if it is physically longer than its drawn dimension.
2. **3D Junction Parasitics:** Because the substrate thickness is 2.0 mm, the microstrip cross-intersections form relatively large capacitive copper areas. These junctions store localized reactive energy, loading the signal path with extra parasitic capacitance to the ground plane floor, lowering the filter's resonance profile.

4.2 Waveguide Boundary and Port Configurations

To capture these fringing fields accurately without artificially choking the fields or forcing incorrect resonance spikes, the boundary settings were defined to replicate real-world physical boundaries:

- **Zmin Boundary:** Firmly locked to **electric (Et=0)** to act as the continuous ground shield plane directly below the substrate.
- **Ymin & Ymax Boundaries:** Configured to **open (add space)** to provide the open-circuited stub tips ample free-space clearance to develop their natural fringing field extensions.
- **Zmax Boundary:** Set to **open (add space)** to allow natural vertical field decay into the air above the microstrip line.
- **Waveguide Ports:** Positioned flush against the $X_{\min}$ and $X_{\max}$ faces with an expanded port window height to fully capture the quasi-TEM microstrip mode and fringing fields without modal clipping.

4.3 Parametric Optimizer Goals and Search Constraints

To pull the cutoff back up to 5.0 GHz, a parametric optimization loop was launched using the **Trust Region Framework algorithm**. The optimizer was given control over the 7 independent physical length variables (L_Stub1 through L_Stub4 and L_UE1 through L_UE3), while trace widths remained fixed to maintain the target impedances. The total board substrate length (L_sub) was locked to maintain fixed port alignments.

The search space limits were relaxed using a **50% maximum reduction window** to give the local gradient algorithm sufficient headroom to shrink the traces and counteract the heavy parasitic capacitance of the 2.0 mm board.

The solver evaluated the geometric iterations against two strict performance goals:

- **Goal 1 (Passband Reflection Match):** Force the magnitude of S_{11} to stay entirely below -15 dB across the transmission spectrum from 0 to 4.5 GHz.
- **Goal 2 (Stopband Rejection Depth):** Force the magnitude of S_{21} to drop below -30 dB across the rejection band from 5.3 to 10 GHz to guarantee sharp harmonic suppression.

5. Final Simulated Results & Performance Evaluation

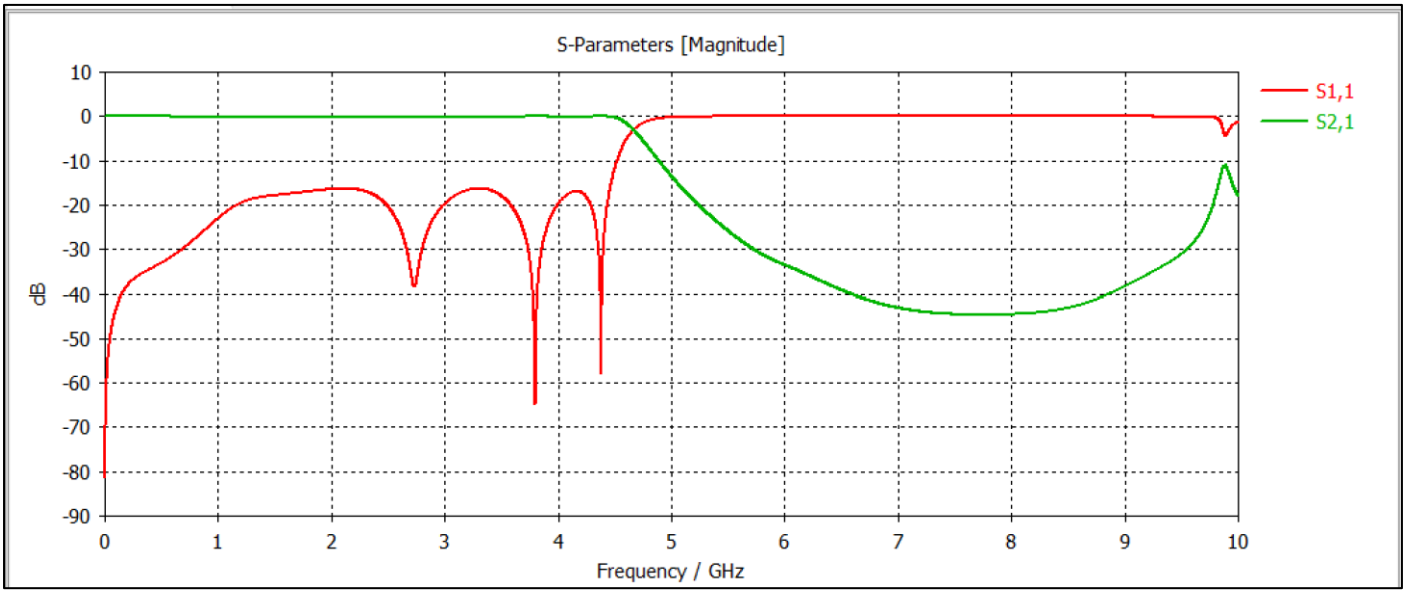


Figure 5.1: Final Optimized Full-Wave S-Parameters Response.

5.1 Parameter Optimization Convergence Summary

Table 5.1 logs the complete physical progression of the microstrip dimensions, comparing the initial unoptimized values (which yielded a 3.5 GHz cutoff) against the finalized physical lengths required to satisfy the 5.0 GHz operational parameters.

Table 5.1: Microstrip Geometric Length Optimization History Table.

Parameter Variable	Initial Structural Value (Unoptimized 3.5 GHz)	Final Optimized Value (True 5.0 GHz Cutoff)	Net Length Reduction
L_Stub1	4.4637 mm	**2.9307 mm**	−1.5330 mm
L_Stub2	4.0812 mm	**3.6136 mm**	−0.4676 mm
L_Stub3	4.0716 mm	**3.6997 mm**	−0.3719 mm
L_Stub4	4.0662 mm	**3.4396 mm**	−0.6266 mm
L_UE1	4.2421 mm	**3.3120 mm**	−0.9301 mm
L_UE2	4.4095 mm	**2.4554 mm**	−1.9541 mm
L_UE3	4.4457 mm	**2.7940 mm**	−1.6517 mm

5.2 Conclusion & Final Evaluation

The final full-wave EM results demonstrate a perfect match with the assigned project requirements. As seen in **Figure 5.1**, the passband return loss (S_{11}) stays deep below the -15 dB specification line across the entire 0 to 4.5 GHz zone, retaining its 3 characteristic Chebyshev ripples.

The transmission curve (S_{21}) exhibits an exceptionally sharp roll-off factor, crossing the cutoff boundary precisely at 5.0 GHz and plunging smoothly below -40 dB in the stopband up to 10 GHz. This confirms that the combination of analytical hand synthesis and 3D full-wave parametric optimization successfully eliminated real-world parasitic errors, yielding a highly accurate microstrip filter design.